

```
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Apply Sobel X
sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=5)

# Convert to 8-bit image
sobel_x = cv2.convertScaleAbs(sobel_x)

# Save the output
cv2.imwrite("sobel_x.jpg", sobel_x)

# Display images
print("Original Image:")
cv2_imshow(img)

print("Sobel X Edge Detection:")
cv2_imshow(sobel_x)
```


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Saving Screenshot 2026-07-22 105943.png to Screenshot 2026-07-22 105943 (1).png

Original Image:

